Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Technology development, technology transfer, and industrialization of ultra-low loading, thin-film catalyst coatings for PEM (and AEM) water electrolysis.*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		Chemicals	
	Medium Category	Small Category	Medium Category	Small Category
	New & Renewable Energy	Hydrogen	Chemical Processes	Catalyst Application Technologies
Concept of Proposed Technology	*Thin-film coatings offer superb coating control, uniformity, and conformality. The research of TNO Holst Centre shows how different thin-film processes applied to PEM electrolyser electrodes and membranes can reduce the platinum and iridium utilization as a catalyst, while simultaneously, if applied with optimally developed processes, can enhance the electrochemical kinetics thus improving conversion efficiency and durability of PEM electrolysers, both critical factors towards reduction of cost of green hydrogen production. This enhancement allows for next generation PEM cells to be manufactured with a fraction of current material costs while simultaneously increasing the electrochemical performance and conversion efficiency.*			

(max 10 pages)

Rationale for Joint R&D Project	*Green hydrogen production commercial viability is limited by the capital and operational costs of transforming renewable electricity into hydrogen molecules. Both should be addressed to create a sound business case. Innovative perspectives such as the integration of novel process technologies, such as thin-film coatings, into established electrochemical processes, are required to reduce costs, expensive material utilization, and improve conversion efficiency further reducing operational cost.* *The technology TNO Holst Centre has developed is innovative and has demonstrated its potential for PEM electrolysis, however it must be upscaled and taken by commercial partners for it to make a societal impact.* *Innovative Korean SMEs have the potential of exploiting and rapidly upscaling this novel technologies based on a combination of their position as new incumbents to the electrolysis market, a strong technological base and key geographical position to service both the European and the growing Asian market.*
Key Words	1. PEM electrolysis 2. Low iridium catalyst coating 3. Spatial atomic layer deposition 4. Thin-film coatings 5. Electrochemical research & development

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)
R&D Objectives	• Development of optimized (for Korean SME specifications) ultra-low iridium coating process for PEM PTLs including but not limited to Spatial Atomic Layer Deposition on industry relevant sizes [1,2,] • Development of surface homogenization process technology to serve as pre-processing for ultra-low iridium coating purposes [3] • Research of catalyst layer and electrode degradation • Manufacture of prototype samples for Korean SME to perform further research and development activities • Development of lifetime enhancing processes for catalyst layer and electrode • Development of protective coatings for PEM catalyst layer and electrodes • Technology transfer of developed processes to Koren SME production team			
R&D Target Specifications	PEM cell performance target: 1.85 V @ 3.0 A/cm^2 Iridium loading target: ranging from a loading of 0.2 mgIrOx/cm^2 in a "high loading" scenario, up to a loading of 0.1 mgIrOx/cm^2 in a "low loading" scenario. Durability target: intermediate target 1,000 hours @ 5% degradation; final target 5,000 hours @ 5% degradation. Sample size: 5x5 cm^2 → 10x10 cm^2 → Towards 16x35 cm^2 by the end of the project.			
R&D Plan	**Phase 1 (M1 - M6)** WP1 (M1 - M2) Target setting; Jointly defining targets, specifications and requirements for Korean SME electrodes and components; Sample exchange WP2 (M2 – M4) Process optimization for ultra-low iridium coated PTLs; A series of samples will be coated, examined by TNO Holst Centre by physical and electrochemical testing, and the process will be adapted to achieve target specifications WP3 (M3 – M6) Development of lifetime enhancing techniques for ultra-low iridium coated PTLs; Through an iterative process, different proven techniques, such as pre and post processing, surface homogenization, interfacial layers, protective layers, etc. will be explored to seek a durability enhancement of the low-iridium coated PTLs WP4 (M5 - M6) Winning sample development; A winning prototype will be manufactured to showcase phase 1 progress and steer development of phase 2 **Go - No Go (M6)** **Phase 2 (M6 – M32)** WP2 (M6 – M30) Process optimization for ultra-low iridium coated PTLs II; A continuation of phase 1 developments. Regular updates will be held. The Korean SME will be invited during the duration of this work package to TNO Holst Centre facilities to help the technology transfer process from an early stage. WP3 (M6 – M30) Development of lifetime enhancing techniques for ultra-low iridium coated PTLs I; a continuation of phase 1 developments. Regular updates will be held. The Korean SME will be			

	invited during the duration of this work package to TNO Holst Centre facilities to help the technology transfer process from an early stage. WP5 (M6 – M36) Prototype manufacture; Prototype samples will be manufactured and shipped to Korean SME for further investigation. WP6 (M24 - M36) Tech Transfer; The TNO team will support the Koren SME in the industrialization process; Visits will be scheduled from the Korean SME to TNO Holst Centre R&D facilities for technology transfer and sharing best practices. Visits will be scheduled from the TNO Holst Centre team to the Korean SME production facilities to commission the transferred technology.

(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Develop the thin-film coating technology optimized for the product of the Korean SME • Develop additional process technologies that guarantee target performance and durability • Perform physical characterization and electrochemical testing to guide the development and optimization process. • Supply the Korean SME with prototype samples to be further investigated by them • Support the Korean SME research and production team to replicate the obtained results in the Korean facilities • Support the Korean SME production team to industrialize the manufacturing process • Update the Koren SME on a regular basis to guarantee a right direction of development
Expected R&R of Korean SME Partner	• Assist the development and optimization process (co-development and co-optimization) for the thin film coating and additional processes by providing requirements and characteristics associated to their unique existing product • Perform lifetime/durability tests on samples supplied by TNO Holst Centre and share the results with TNO Holst Centre • Host the TNO Holst Centre at their facilities to facilitate the tech transfer from TNO Holst Centre to the Korean SME • Visit the TNO Holst Centre facilities to facilitate the tech transfer from TNO Holst Centre to the Korean SME •

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	*The full project is expected to deliver a first of it's kind coating PEM cell that outperforms conversion efficiencies of state of the art PEM cells, at a fraction of its material cost.* *The prototype samples that will be provided as an outcome of this project will pave the way for the Korean SME to engage prospective customers and/or investors, and help commercialize their product, bringing the benefits to society.* *Once a fully mature product is developed and successfully industrialized, this will bear the potential to disrupt state of the art electrolyser technologies.*

6. References

(1) Poodt P.W.G, Balder J.E., "METHOD FOR PROVIDING A SUBSTRATE FOR AN ELECTROCHEMICAL CELL WITH A CATALYTIC MATERIAL", 2022, WO2022015161A1 (link)

(2) Shirvanian A.P., "THIN FILM POROUS CATALYST SHEET", 2023, WO2023287284A1 (link)

(3) Diaz-Morales O., Shen J., Santana C.S., Bens F., van Dijk B., "Spatial Atomic Layer Deposition to Manufacture Catalyst Coated Electrodes for PEM Water Electrolysis with Ultra-Low Loadings of Iridium", ECS Meeting Abstracts (2024) MA2024-02 2842 (link)

(max 10 pages)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization